CSA0711 Computer Networks

Assignment

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Assignment Title	Analyze Protocol Behaviour and Measure Network Performance Across Alternative Topologies Using Simulation
Course Outcome(s) – CO	· CO3: Analyze and compare the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions. · CO4: Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards · CO5: Measure network performance using key metrics with simulation tools
Bloom's Taxonomy Level	L5 – Evaluate; L6 – Create
SDG Mapping	SDG 9 – Industry, Innovation and Infrastructure;

1. Develop, implement/simulate, and evaluate a campus or enterprise network in which different applications generate different traffic characteristics. The network shall contain at least four interconnected segments and a minimum of 40 end devices. Design four alternative network structures—star, mesh, bus, and hybrid—using appropriate networking devices, transmission media, and cabling standards. Implement the selected network design in a suitable simulation tool such as Packet Tracer, GNS3, EVE-NG, or equivalent. Conduct controlled experiments to analyze TCP, UDP, HTTP, and DNS under at least three varying network conditions such as low traffic, high traffic, increased delay, packet loss, or link congestion. Measure a minimum of five key network performance metrics, including appropriate combinations of latency, throughput, packet loss, delay, jitter, utilization, and successful delivery rate. Repeat tests where appropriate, organize the measured data in tables and graphs, compare protocol and topology performance, identify bottlenecks, and justify the final topology and engineering decisions based on measured simulation results.

Detailed Assessment Rubric with Performance Levels (Mapped to Assignment Requirements)

Assessment Criterion	**Marks**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
1. Problem Understanding & Network Requirements	10	Clearly identifies application needs, traffic patterns, topology constraints, device/media requirements, performance	Identifies most requirements with minor omissions.	Identifies basic requirements but lacks depth.	Requirements are incomplete or poorly understood.

		objectives and test conditions.			
2. Protocol Performance Analysis	20	Systematically analyzes TCP, UDP, HTTP and DNS under multiple controlled conditions and explains differences using quantitative and qualitative evidence.	Provides a sound comparison with adequate evidence.	Provides a basic comparison with limited evidence.	Analysis is inaccurate, superficial, or unsupported.
3. Topology Design & Network Infrastructure	15	Creates technically correct star, mesh, bus and hybrid topologies and selects appropriate networking devices, media and cabling standards with clear justification.	Most topology and infrastructure choices are appropriate.	Basic designs are present but contain gaps or weak justification.	Designs are incomplete or technically unsuitable.
4. Simulation Implementation & Experiment Design	15	Builds the proposed network and controlled experiments in a suitable simulator; test cases are repeatable, clearly defined and technically valid.	Implementation and experiments are mostly correct.	Simulation is functional but experiments are limited or poorly controlled.	Simulation or experiment design is incomplete.
5. Network Performance Measurement	15	Accurately measures multiple metrics such as latency, throughput, packet loss, jitter, delay and/or delivery	Measures appropriate metrics with adequate accuracy.	Measures a limited set of metrics or has weak measurement methodology.	Performance metrics are missing, inaccurate, or not reproducible.

		rate; results are organized and repeatable.			
6. Results Interpretation & Engineering Decisions	10	Uses tables/graphs to interpret results, identifies bottlenecks and trade-offs, and recommends topology/protocol choices based on measured evidence.	Interprets results and gives reasonable recommendations.	Provides limited interpretation or weak recommendations.	Conclusions are unsupported by measured results.
7. Broader Considerations & Professional Responsibility	5	Considers reliability, scalability, standards, efficient resource use, responsible experimentation, and practical network deployment considerations.	Addresses most broader considerations adequately.	Limited discussion of broader design or professional considerations.	Negligible consideration of broader impacts.
8. Technical Documentation & Reflection	10	Provides a professional report with diagrams, methodology, configurations, raw/processed results, limitations, references and a thoughtful 300–500 word reflection.	Complete report and reflection with good clarity.	Documentation or reflection lacks depth.	Poor, incomplete, or missing documentation/reflection.

TOTAL 100

Deliverables

To receive full credit, each submission must include the following deliverables:

1. Pseudo code

2. Implementation & Results
3. Github Upload

1. Problem Statement

Modern campus and enterprise networks support many applications such as web browsing, file transfer, DNS services, video communication and real-time applications. These applications generate different traffic patterns and therefore place different demands on a network. TCP-based applications prioritize reliable delivery, while UDP-based applications generally prioritize speed and low overhead. HTTP commonly transfers larger amounts of data, whereas DNS normally uses small request-response messages.

This project develops and evaluates a simulated enterprise network containing four interconnected segments and at least 40 end devices. Four alternative network structures—Star, Mesh, Bus and Hybrid—are designed and compared. The selected design is implemented in Cisco Packet Tracer. Controlled experiments are conducted using TCP, UDP, HTTP and DNS traffic under low traffic, high traffic, increased delay and packet-loss conditions.

Performance is evaluated using latency, throughput, packet loss, jitter, network utilization and successful delivery rate. Results are organized into tables and graphs to identify bottlenecks and justify the final topology and engineering decisions.

2. Objectives

Design an enterprise network with at least four interconnected segments.

Connect a minimum of 40 end devices.

Develop Star, Mesh, Bus and Hybrid topology alternatives.

Select suitable routers, switches, transmission media and cabling standards.

Implement the selected design in Cisco Packet Tracer.

Generate and evaluate TCP, UDP, HTTP and DNS traffic.

Test the network under multiple controlled conditions.

Measure latency, throughput, packet loss, jitter, utilization and delivery rate.

Compare protocol and topology performance using measured evidence.

Identify bottlenecks and recommend an appropriate enterprise topology.

3. Requirements and Environment Used

3.1 Software Requirements

Component	Requirement
Simulation Tool	Cisco Packet Tracer
Operating System	Windows / Linux
Network Configuration	Cisco IOS
Analysis Tools	Microsoft Excel / Python
Documentation	Microsoft Word

3.2 Network Devices

Routers

Layer 2 or Layer 3 switches

40 or more PCs/end devices

DNS server

HTTP/Web server

Optional wireless access points

3.3 Transmission Media and Cabling

Copper straight-through Ethernet cable for end-device to switch connections.

Copper crossover cable where directly connecting similar legacy Ethernet devices, if required by the simulator.

Fiber-optic cable for high-speed backbone links where supported.

Serial links may be used for WAN-style router connections if required.

4. Network Segmentation

Segment	Purpose	End Devices
Segment 1	Administration	10
Segment 2	Academic / Operations	10

Segment 3	Research / Development	10
Segment 4	IT / Server Services	10
Total	Enterprise Network	40

5. Alternative Network Topologies

5.1 Star Topology

All end devices are connected to a central switch.

Conceptual layout: PC — Switch — PC, with additional PCs connected to the same central switch.

Easy configuration and troubleshooting.

Simple expansion.

Failure of one access cable normally affects only one device.

Central switch can become a critical point of failure or congestion.

5.2 Mesh Topology

Core network devices have multiple connections to other network devices, creating alternative paths.

High reliability and fault tolerance.

Multiple paths can be used when a link fails.

High cabling, port and configuration requirements.

More difficult to scale and manage.

5.3 Bus Topology

Devices share a common backbone in the traditional bus model.

Simple conceptual structure.

Low initial infrastructure requirement.

Shared medium can cause congestion.

A backbone failure can affect many or all devices.

Poor scalability for modern enterprise environments.

Implementation note: modern Cisco Packet Tracer environments are primarily based on switched Ethernet and do not provide a practical modern Ethernet bus equivalent to historical coaxial Ethernet. The Bus topology should therefore be documented as a conceptual/reference design or simulated only where the chosen simulator supports a shared-medium representation.

5.4 Hybrid Topology

The proposed hybrid design combines Star access networks with partial Mesh redundancy at the core.

Good balance of reliability, scalability and manageability.

Allows redundant backbone paths.

Supports departmental separation.

More practical for enterprise deployment than a complete mesh.

Requires more planning than a simple star.

6. Selected Topology

The Hybrid Star + Partial Mesh topology is selected as the final design. A complete Mesh provides excellent redundancy but increases cost and complexity. A simple Star is easier to manage but depends heavily on central infrastructure. A Bus design is not well suited to a modern enterprise network. The Hybrid approach provides a practical balance between performance, reliability, scalability, cost and maintainability.

7. Proposed IP Addressing Scheme

Network Segment	Network Address	Default Gateway
Administration	192.168.10.0/24	192.168.10.1
Academic / Operations	192.168.20.0/24	192.168.20.1
Research / Development	192.168.30.0/24	192.168.30.1
IT / Servers	192.168.40.0/24	192.168.40.1

Example server address: 192.168.40.10. Example HTTP/DNS service host: 192.168.40.20.

8. Application Traffic Model

Application	Protocol	Traffic Characteristics
Web browsing	TCP / HTTP	Reliable, bursty and moderate-to-large data transfer
DNS queries	UDP / TCP	Small request-response messages
File transfer	TCP	Large reliable data transfer
Real-time traffic simulation	UDP	Delay and jitter sensitive

Connectivity testing	ICMP	Small packets used for latency testing

9. Experimental Conditions

Condition	**Description**	**Expected Effect**
Low Traffic	Small number of devices generate traffic	Low latency and utilization
High Traffic	Many devices transmit simultaneously	Higher utilization and possible congestion
Increased Delay	Additional delay introduced on a link	Higher response time and latency
Packet Loss	Loss introduced on a selected link	Reduced delivery rate and possible retransmissions

10. Performance Metrics

10.1 Latency

Latency is the time required for a packet to travel from source to destination. Lower latency indicates better responsiveness.

10.2 Throughput

Throughput is the amount of successfully transmitted data per unit time. Higher throughput indicates better effective network capacity.

10.3 Packet Loss

Packet loss percentage can be calculated as: (Lost Packets / Sent Packets) × 100. Lower packet loss is preferred.

10.4 Jitter

Jitter represents variation in packet delay. Low jitter is particularly important for real-time UDP traffic.

10.5 Successful Delivery Rate

Delivery Rate (%) = (Received Packets / Sent Packets) × 100.

10.6 Network Utilization

Utilization indicates how much of available network capacity is being used. Very high utilization can indicate congestion.

11. Pseudo Code

```
BEGIN

    Initialize network simulator

    Create four enterprise network segments
    Create minimum 40 end devices
    Add routers, switches and servers

    Design Star topology
    Design Mesh topology
    Design Bus topology
    Design Hybrid topology

    Configure IP addressing
    Configure routing between network segments
    Configure DNS service
    Configure HTTP service

    Verify basic connectivity using ICMP

    FOR each topology

        Generate TCP traffic
        Generate UDP traffic
        Generate HTTP traffic
        Generate DNS traffic

        FOR each network condition

            Apply low-traffic condition
            Measure latency, throughput, packet loss,
            jitter, utilization and delivery rate
            Store measurements

            Apply high-traffic condition
            Repeat measurements
            Store measurements
```

```
        Apply increased-delay condition
        Repeat measurements
        Store measurements

        Apply packet-loss condition
        Repeat measurements
        Store measurements

      END FOR

   END FOR

   Compare topology performance
   Compare protocol performance
   Identify bottlenecks
   Generate tables and graphs
   Select final topology using measured evidence

END
```

12. Protocol Testing Pseudo Code

```
BEGIN

   FOR each protocol in TCP, UDP, HTTP, DNS

      Configure source and destination
      Generate protocol traffic
      Capture transmitted packets
      Capture received packets

      Calculate latency
      Calculate throughput
      Calculate packet loss
      Calculate jitter where applicable
      Calculate successful delivery rate

      Store results

   END FOR

   Compare protocol performance
```

END

13. Implementation

Open Cisco Packet Tracer and create the enterprise workspace.

Create four network segments for Administration, Academic/Operations, Research/Development and IT/Servers.

Add routers, switches, 40 end devices and application servers.

Connect access devices using appropriate Ethernet cabling.

Connect the core using high-speed links and configure redundant paths for the hybrid design.

Assign IP addresses and default gateways to all devices.

Configure routing between the four network segments.

Configure DNS records and HTTP service on the server segment.

Verify connectivity using ping and application tests.

Run the controlled traffic experiments and record measurements.

14. Routing Configuration

Static routing, RIP or OSPF can be used. OSPF is recommended for the enterprise-style implementation because it dynamically learns routes and is more appropriate for an interconnected multi-segment network.

15. DNS and HTTP Implementation

DNS

Configure a DNS service in the IT/Server segment. Example logical record: www.enterprise.local → 192.168.40.20.

HTTP

Enable HTTP on the web server and configure a simple webpage. Clients can request the page using the configured DNS name.

16. Connectivity Verification

Ping between devices in different departments.

Ping from clients to the DNS server.

Resolve the configured enterprise hostname.

Access the HTTP server from multiple segments.

Confirm that routing tables contain paths to all four networks.

17. Implementation & Results

This section is intended to contain the actual measurements obtained from the simulation. The values should be copied from Packet Tracer, GNS3, EVE-NG or the selected equivalent rather than invented. Repeat each important test to improve reliability.

Test	Protocol	Condition	Latency	Throughput	Packet Loss	Jitter	Delivery Rate
1	TCP	Low Traffic	Record	Record	Record	Record	Record
2	UDP	Low Traffic	Record	Record	Record	Record	Record
3	HTTP	Low Traffic	Record	Record	Record	Record	Record
4	DNS	Low Traffic	Record	Record	Record	Record	Record
5	TCP	High Traffic	Record	Record	Record	Record	Record
6	UDP	High Traffic	Record	Record	Record	Record	Record
7	HTTP	High Traffic	Record	Record	Record	Record	Record
8	DNS	High Traffic	Record	Record	Record	Record	Record
9	TCP	Increased Delay	Record	Record	Record	Record	Record
10	UDP	Increased Delay	Record	Record	Record	Record	Record
11	HTTP	Packet Loss	Record	Record	Record	Record	Record
12	DNS	Packet Loss	Record	Record	Record	Record	Record

18. Topology Comparison Results

Topology	Latency	Throughput	Packet Loss	Jitter	Reliability	Scalability
Star	Measured	Measured	Measured	Measured	Good	Good
Mesh	Measured	Measured	Measured	Measured	Very Good	Moderate
Bus	Measured	Measured	Measured	Measured	Low	Low
Hybrid	Measured	Measured	Measured	Measured	Very Good	Very Good

19. Recommended Graphs

Latency versus network condition.

Throughput versus network condition.

Packet loss versus network condition.

Jitter versus protocol.

Successful delivery rate versus topology.

Network utilization versus traffic load.

20. Protocol Performance Analysis

TCP

TCP provides reliable delivery through acknowledgements, sequencing and retransmission. When packet loss or congestion increases, retransmissions and congestion control can increase effective delay and reduce useful throughput.

UDP

UDP has lower protocol overhead and does not provide built-in retransmission. It can therefore be efficient for delay-sensitive applications, but packet loss can directly reduce successful delivery.

HTTP

HTTP normally operates over TCP. Its performance is therefore affected by TCP connection behavior, latency, congestion and available bandwidth.

DNS

DNS generally produces small request-response messages. Its bandwidth requirement is low, but increased network delay can noticeably affect name-resolution response time.

21. Bottleneck Identification

Overloaded central switches or core links.

Low-bandwidth backbone connections.

Large numbers of simultaneous flows.

Single links connecting multiple network segments.

Packet loss on congested links.

Excessively high network utilization.

A link should be identified as a bottleneck when measurements show consistently high utilization together with increasing latency, reduced throughput, packet loss or jitter under heavier traffic.

22. Engineering Decisions

The final topology should be selected using the measured results rather than assumptions alone. If the hybrid design demonstrates a strong balance of throughput, latency, reliability and scalability, it is recommended for the enterprise network. The decision should also consider cost, cabling requirements, fault tolerance, maintainability and future expansion.

23. Broader Considerations

Reliability: redundant core links can improve service continuity.

Scalability: departmental segments can be expanded without redesigning the entire network.

Standards: use standard Ethernet cabling, IP addressing and routing practices.

Efficient resource use: avoid unnecessary full-mesh links where they provide little benefit.

Responsible experimentation: change one major test condition at a time and document configurations.

Practical deployment: consider equipment cost, maintenance, security and future bandwidth needs.

24. Limitations

Packet Tracer is a simulation and does not reproduce every physical behavior of a real network.

Simulated application traffic may differ from real enterprise traffic.

Hardware processing characteristics may not be fully represented.

Traditional Bus topology is difficult to reproduce in modern switched Ethernet simulation.

Results depend on the simulator model, link speeds and experiment configuration.

25. Conclusion

This project develops and evaluates a simulated enterprise network containing four interconnected segments and at least 40 end devices. Star, Mesh, Bus and Hybrid alternatives are considered, while the selected design is implemented in a network simulator. TCP, UDP, HTTP and DNS traffic are evaluated under different network conditions including low traffic, high traffic, increased delay and packet loss.

Latency, throughput, packet loss, jitter, utilization and successful delivery rate are used to compare performance. Tables and graphs provide quantitative evidence for identifying congestion and bottlenecks. The final topology is selected by balancing measured performance with reliability, scalability, cost and maintainability. The Hybrid Star + Partial Mesh design is proposed as a practical enterprise solution, subject to confirmation by the actual simulation results.